

RPMBT XXIII – Quantum Simulation School

	9 Sept	10 Sept	11 Sept
8:30-9:00	Registration		
9:00-9:30	Welcome+Sponsors	THEORY 3 VQE, intro to errors and their mitigation	THEORY 5 ancilla-free spectral function, circuit knitting
9:30-10:30	THEORY 1 Quantum Simulations		
10:30-11:00	Break	Break	Break
11:00-12:30	EXERCISE 1 Time evolution for Ising chain	EXERCISE 3 VQE and read-out error mitigation	THEORY 6 advanced error mitigation, LCU and QSP
12:30-14:00	Lunch	Lunch	Lunch
14:00-14:20	THEORY 2 Trotter-Suzuki decomposition, linear response	LINKS Foundation	EXERCISE 5 full protocol implementation on hardware
14:20-15:30		THEORY 4 error mitigation for quantum gates	
15:30-16:00	Break	Break	Break
16:00-17:30	EXERCISE 2 Correlation functions and execution on emulators	EXERCISE 4 Execution on quantum hardware with full mitigation	EXERCISE 6 full protocol implementation on hardware, extensions

RPMBT XXIII — Quantum Simulation School

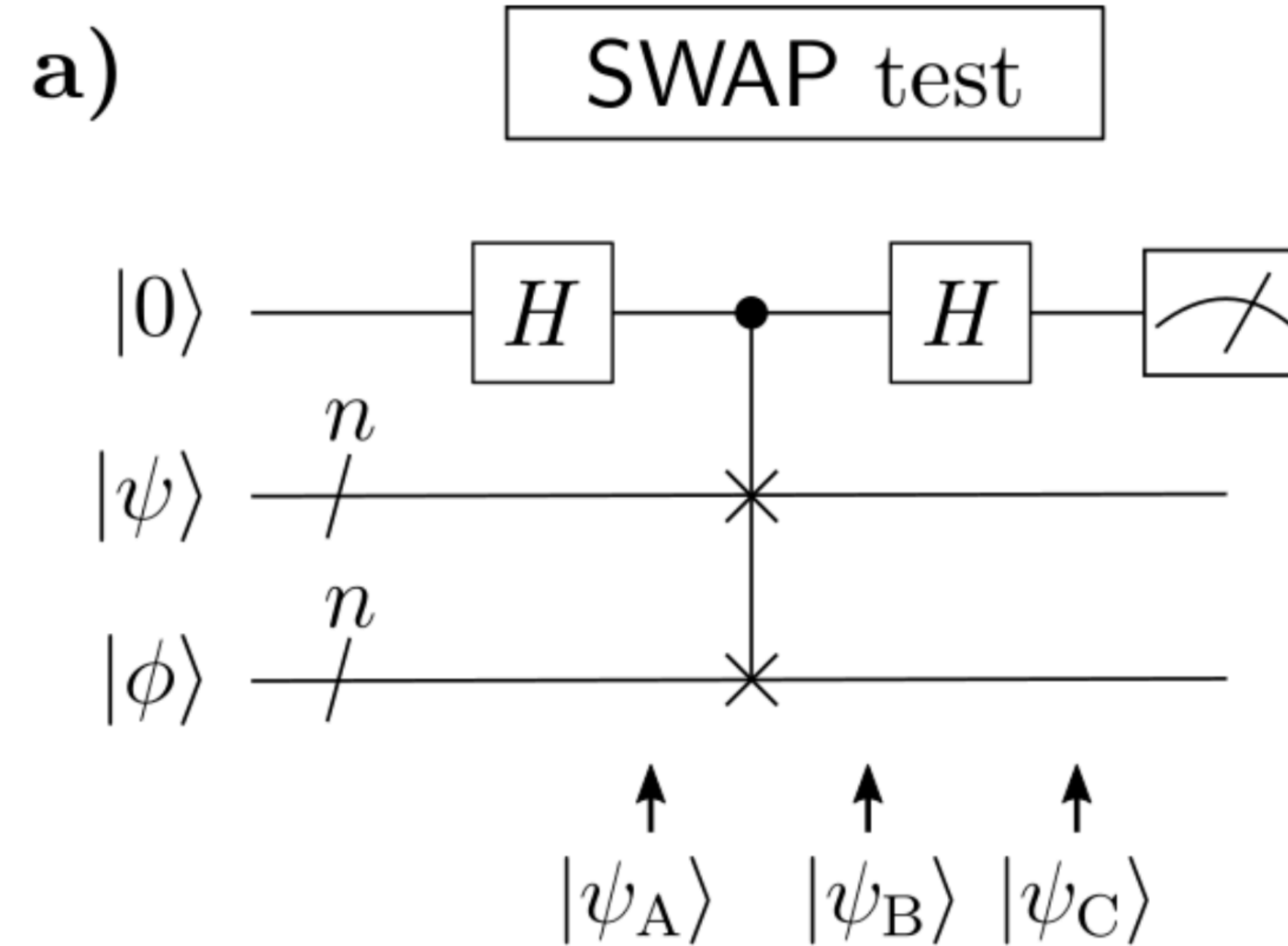
- SWAP test
- Hadamard test
- Iterative phase estimation
- Impact of decoherence
- Bypassing the Hadamard test
 - symmetries
 - mid-circuit measurements

SWAP test

Goal: measure the fidelity between two quantum states

$$F = |\langle \psi | \phi \rangle|^2$$

Quantum circuit



SWAP test

Initial state

$$|0, \psi, \phi\rangle$$

After Hadamard gate

$$\frac{1}{\sqrt{2}} |0, \psi, \phi\rangle + \frac{1}{\sqrt{2}} |1, \psi, \phi\rangle$$

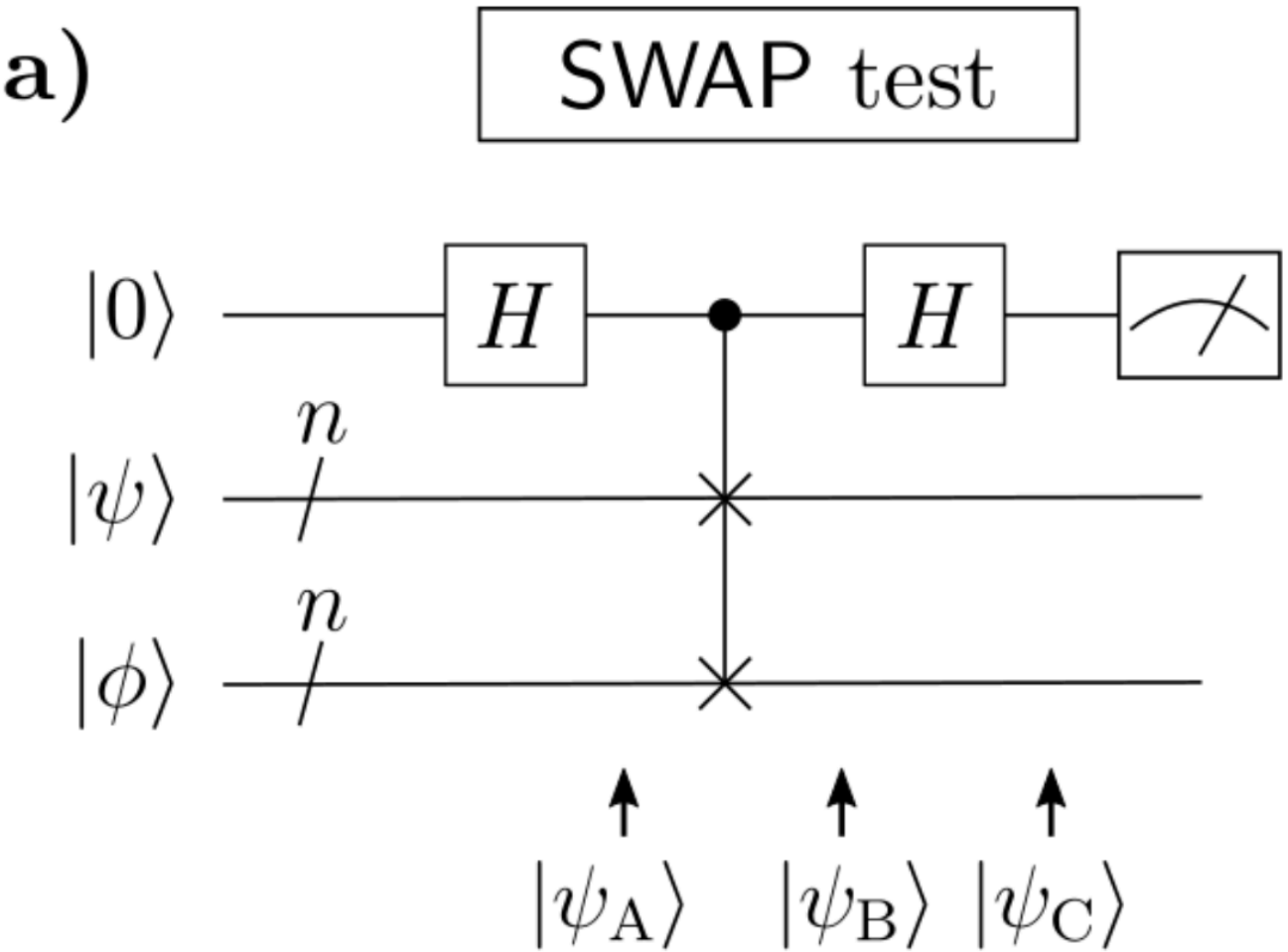
After controlled SWAP gate

$$\frac{1}{\sqrt{2}} |0, \psi, \phi\rangle + \frac{1}{\sqrt{2}} |1, \phi, \psi\rangle$$

After second Hadamard gate

$$\frac{|0\rangle + |1\rangle}{2} |\psi, \phi\rangle + \frac{|0\rangle - |1\rangle}{2} |\phi, \psi\rangle$$

$$\frac{1}{2} |0\rangle [|\psi, \phi\rangle + |\phi, \psi\rangle] + \frac{1}{2} |1\rangle [|\psi, \phi\rangle - |\phi, \psi\rangle]$$



SWAP test

After second Hadamard gate

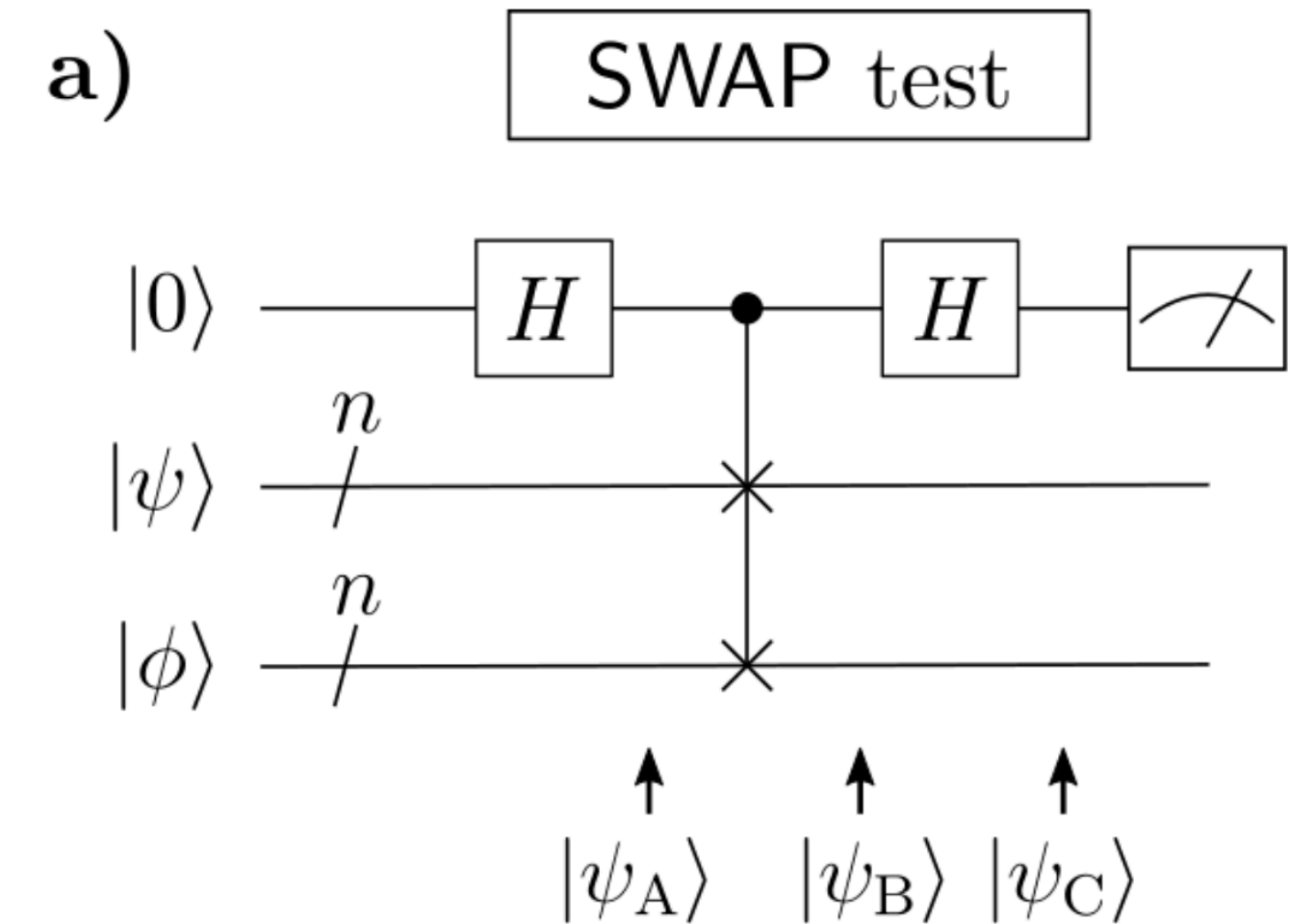
$$\frac{1}{2} |0\rangle [|\psi, \phi\rangle + |\phi, \psi\rangle] + \frac{1}{2} |1\rangle [|\psi, \phi\rangle - |\phi, \psi\rangle]$$

Ancilla measurement

$$p(0) = \frac{1}{4} [\langle\psi, \phi| + \langle\phi, \psi|] [|\psi, \phi\rangle + |\phi, \psi\rangle] = \frac{2 + 2|\langle\psi|\phi\rangle|^2}{4}$$

From $p(0)$ one can estimate the fidelity as

$$2p(0) - 1 = |\langle\psi|\phi\rangle|^2$$



Main source of cost: ancilla and controlled SWAP gates

RPMBT XXIII — Quantum Simulation School

- SWAP test
- Hadamard test
- Iterative phase estimation
- Impact of decoherence
- Bypassing the Hadamard test
 - symmetries
 - mid-circuit measurements

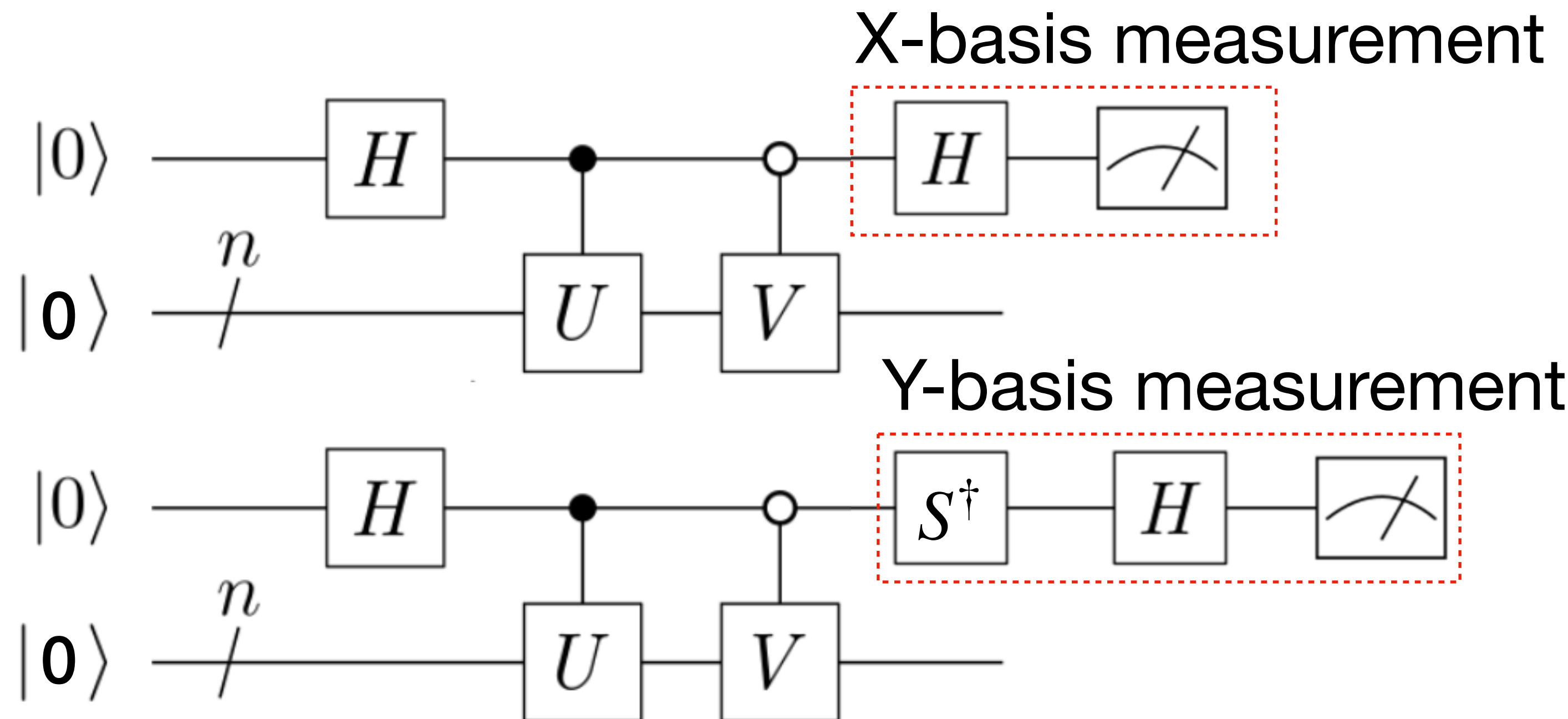
Hadamard test

Goal: compute the inner product between two states prepared by quantum circuits

$$\omega = \langle \psi | \phi \rangle = \langle 0 | V^\dagger U | 0 \rangle$$

Quantum circuit

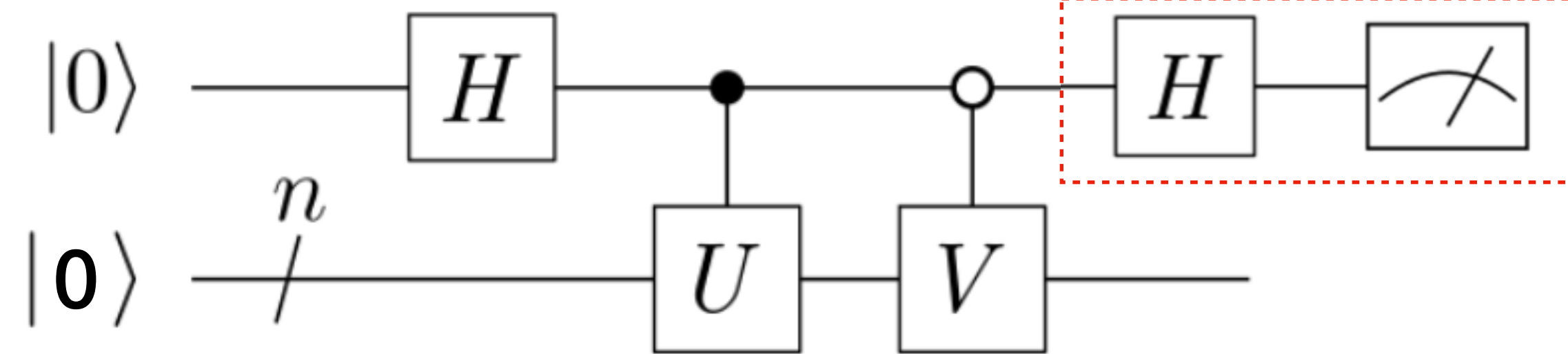
Real part



Imaginary part

Hadamard test

X-basis measurement



Initial state

$$|0, \mathbf{0}\rangle$$

State after Hadamard

$$\frac{|0, \mathbf{0}\rangle + |1, \mathbf{0}\rangle}{\sqrt{2}}$$

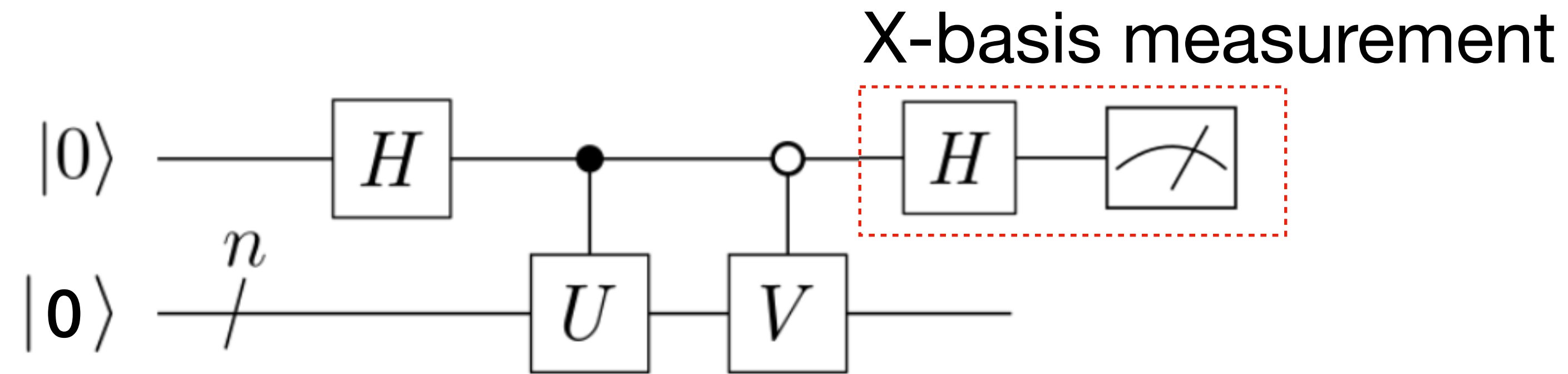
State after controlled unitaries

$$\frac{|0\rangle U|\mathbf{0}\rangle + |1\rangle V|\mathbf{0}\rangle}{\sqrt{2}} = \frac{|0\rangle |\psi\rangle + |1\rangle |\phi\rangle}{\sqrt{2}}$$

State after second Hadamard

$$\frac{|0\rangle + |1\rangle}{2} |\psi\rangle + \frac{|0\rangle - |1\rangle}{2} |\phi\rangle = |0\rangle \frac{|\psi\rangle + |\phi\rangle}{2} + |1\rangle \frac{|\psi\rangle - |\phi\rangle}{2}$$

Hadamard test



State after second Hadamard

$$\frac{|0\rangle + |1\rangle}{2} |\psi\rangle + \frac{|0\rangle - |1\rangle}{2} |\phi\rangle = |0\rangle \frac{|\psi\rangle + |\phi\rangle}{2} + |1\rangle \frac{|\psi\rangle - |\phi\rangle}{2}$$

Measurement

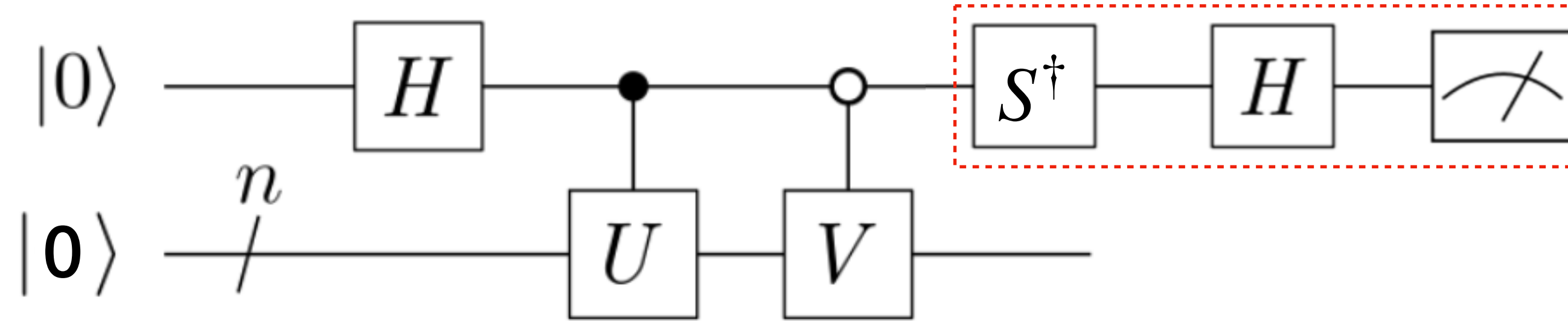
$$p(0) = \left\| \frac{|\psi\rangle + |\phi\rangle}{2} \right\|^2 = \frac{\langle\psi|\psi\rangle + \langle\psi|\phi\rangle + \langle\phi|\psi\rangle + \langle\phi|\phi\rangle}{4} = \frac{2 + 2\text{Re}[\omega]}{4}$$

$$p(1) = \left\| \frac{|\psi\rangle - |\phi\rangle}{2} \right\|^2 = \frac{\langle\psi|\psi\rangle - \langle\psi|\phi\rangle + \langle\phi|\psi\rangle - \langle\phi|\phi\rangle}{4} = \frac{2 - 2\text{Re}[\omega]}{4}$$

$$p(0) - p(1) = \text{Re}[\omega]$$

Hadamard test

Y-basis measurement



Initial state

$$|0, \mathbf{0}\rangle$$

State after Hadamard

$$\frac{|0, \mathbf{0}\rangle + |1, \mathbf{0}\rangle}{\sqrt{2}}$$

State after controlled unitaries

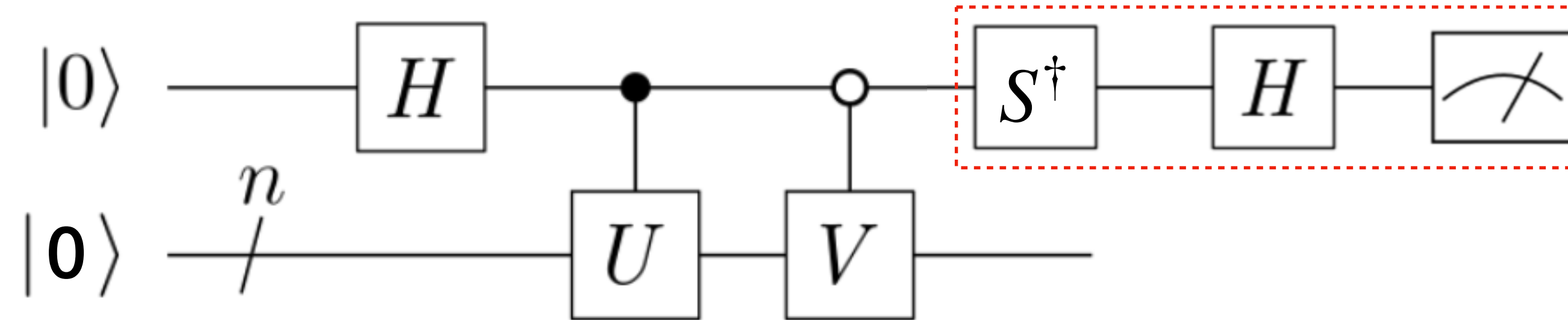
$$\frac{|0\rangle U|\mathbf{0}\rangle + |1\rangle V|\mathbf{0}\rangle}{\sqrt{2}} = \frac{|0\rangle |\psi\rangle + |1\rangle |\phi\rangle}{\sqrt{2}}$$

State after $S^\dagger$

$$\frac{|0\rangle |\psi\rangle - i |1\rangle |\phi\rangle}{\sqrt{2}}$$

Hadamard test

Y-basis measurement



State after second Hadamard

$$\frac{|0\rangle + |1\rangle}{2} |\psi\rangle - i \frac{|0\rangle - |1\rangle}{2} |\phi\rangle = |0\rangle \frac{|\psi\rangle - i|\phi\rangle}{2} + |1\rangle \frac{|\psi\rangle + i|\phi\rangle}{2}$$

Measurement

$$p(0) = \left\| \frac{|\psi\rangle - i|\phi\rangle}{2} \right\|^2 = \frac{\langle\psi|\psi\rangle - i\langle\psi|\phi\rangle + i\langle\phi|\psi\rangle + \langle\phi|\phi\rangle}{4} = \frac{2 + 2\text{Im}[\omega]}{4}$$

$$p(1) = \left\| \frac{|\psi\rangle + i|\phi\rangle}{2} \right\|^2 = \frac{\langle\psi|\psi\rangle + i\langle\psi|\phi\rangle - i\langle\phi|\psi\rangle + \langle\phi|\phi\rangle}{4} = \frac{2 - 2\text{Im}[\omega]}{4}$$

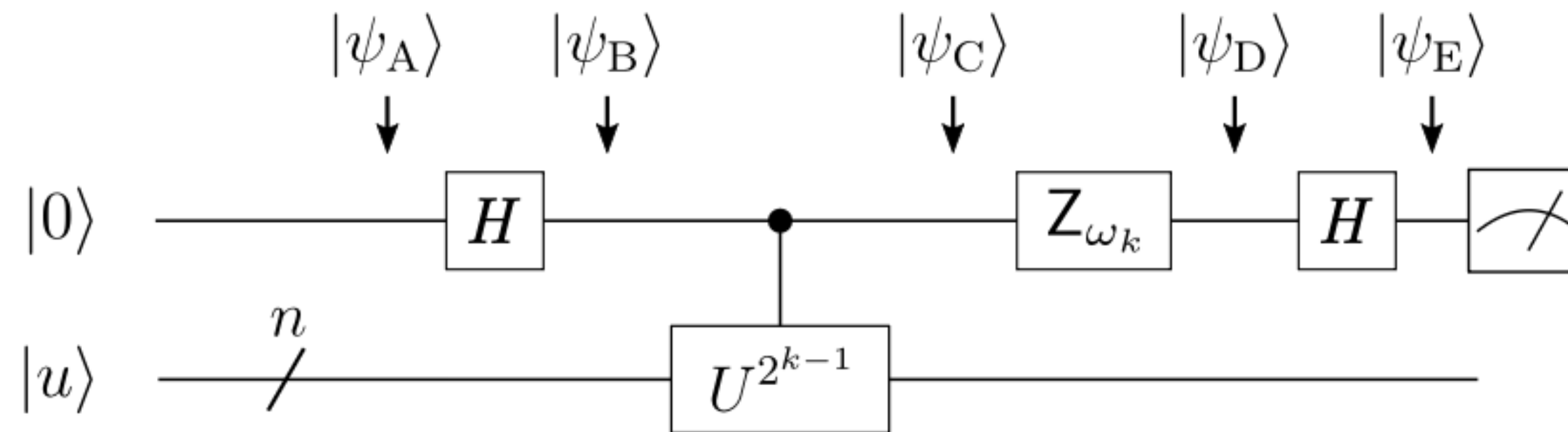
$$p(0) - p(1) = \text{Im}[\omega]$$

RPMBT XXIII — Quantum Simulation School

- SWAP test
- Hadamard test
- Iterative phase estimation
- Impact of decoherence
- Bypassing the Hadamard test
 - symmetries
 - mid-circuit measurements

Iterative Phase Estimation

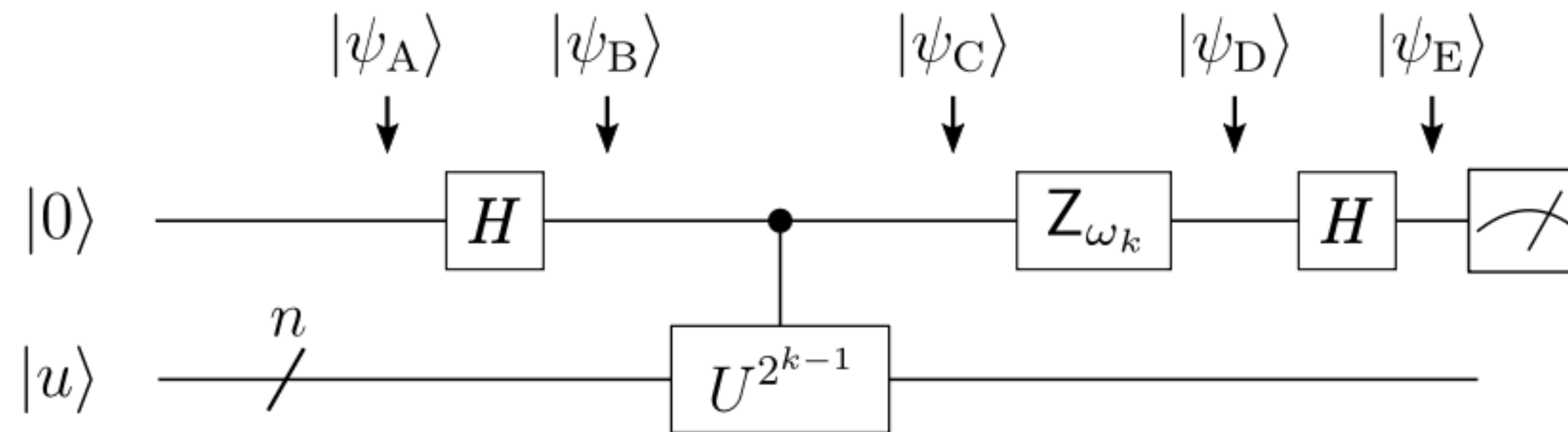
GOAL: given a register of qubits prepared in an eigenstate $|u\rangle$ of a unitary U with unknown eigenvalue, find the eigenvalue



Why bother? Suppose we have prepared the ground state of a Hamiltonian H . Then, this state is also an eigenfunction of $U = \exp(-itH)$. From the eigenvalues of U we can read the ground-state energy of the system.

Iterative Phase Estimation

GOAL: given a register of qubits prepared in an eigenstate $|u\rangle$ of a unitary U with unknown eigenvalue, find the eigenvalue

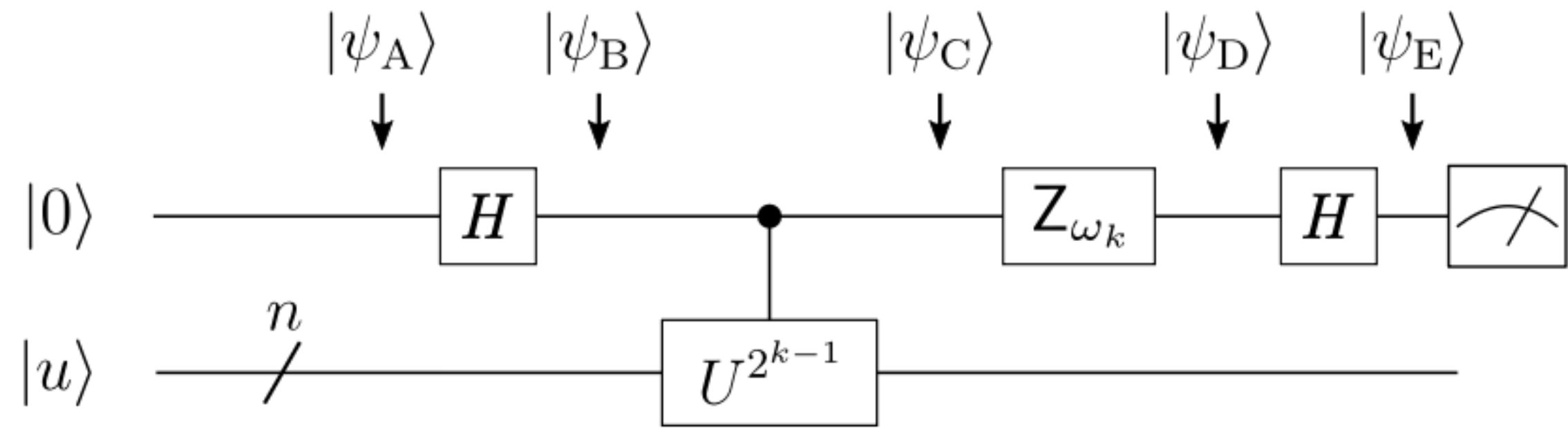


Writing the eigenvalue as $U|u\rangle = e^{i2\pi\phi}|u\rangle$ with

$$\phi = \sum_{j=1}^t \frac{\phi_j}{2^j}, \quad \text{where } \phi_1, \dots, \phi_t \in \{0, 1\}.$$

the goal is to find the binary digits ϕ_j

Iterative Phase Estimation



State A

$$|\psi_A\rangle = |0\rangle|u\rangle$$

State B

$$|\psi_B\rangle = \frac{1}{\sqrt{2}}[|0\rangle|u\rangle + |1\rangle|u\rangle]$$

State C

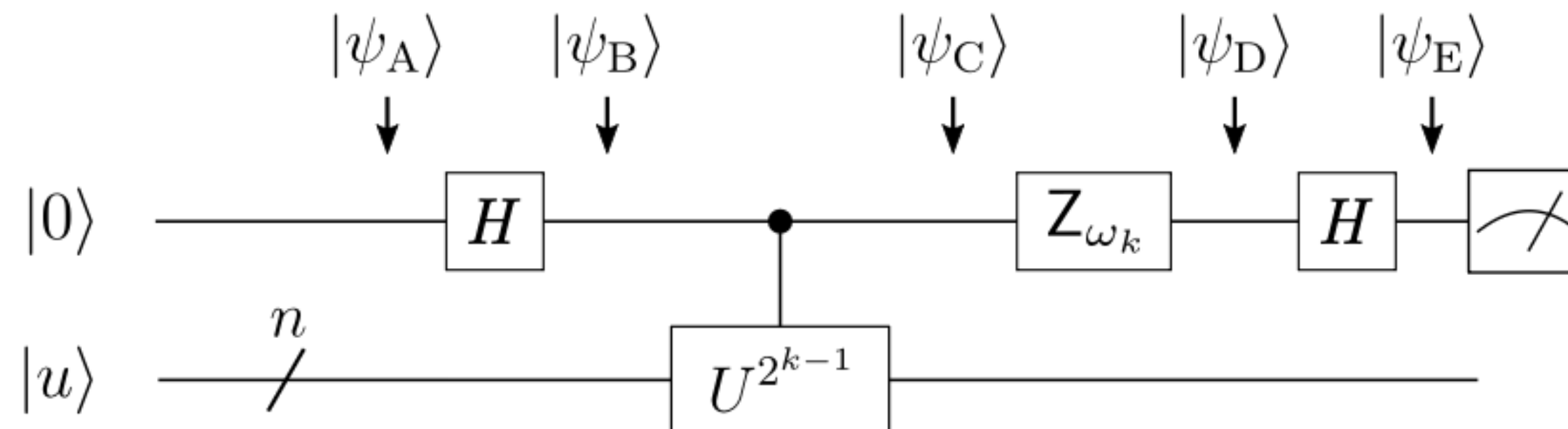
$$|\psi_C\rangle = \frac{1}{\sqrt{2}}[|0\rangle|u\rangle + e^{i2\pi\phi2^{k-1}}|1\rangle|u\rangle]$$

State D

$$|\psi_D\rangle = \frac{1}{\sqrt{2}}[|0\rangle + e^{i(\omega_k+2\pi\phi2^{k-1})}|1\rangle]|u\rangle = \frac{1}{\sqrt{2}}[|0\rangle + e^{i2\gamma}|1\rangle]|u\rangle]$$

$$\gamma = \omega_k/2 + \pi\phi2^{k-1}$$

Iterative Phase Estimation



State E

$$|\psi_E\rangle = \left[\frac{1 + e^{i2\gamma}}{2} |0\rangle + \frac{1 - e^{i2\gamma}}{2} |1\rangle \right] |u\rangle$$

$$p(0) = \left| \frac{1 + e^{i2\gamma}}{2} \right|^2 = \cos^2 \gamma, \quad p(1) = \left| \frac{1 - e^{i2\gamma}}{2} \right|^2 = \sin^2 \gamma$$

Several iterations of the quantum circuits are run with $t = k \rightarrow t = 1$.

The first iteration ($t = k, \omega_k = \omega_t = 0$) finds ϕ_t because

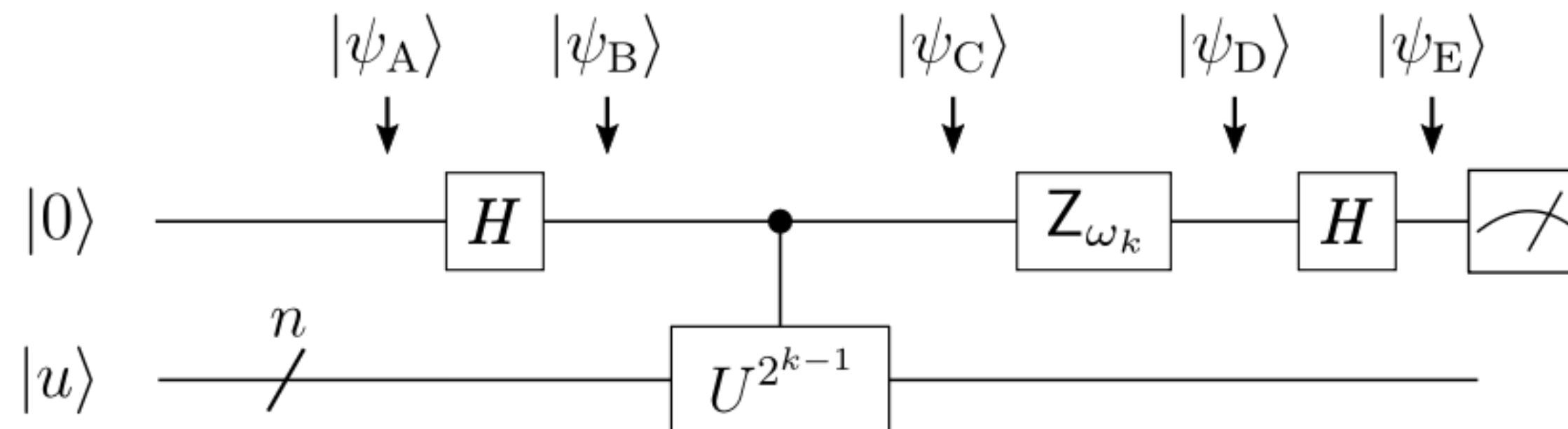
$$\gamma = \frac{\omega_k}{2} + \pi\phi 2^{k-1} = \pi\phi 2^{t-1} = \pi \underbrace{\sum_{j=1}^{t-1} \frac{\phi_j}{2^j} 2^{t-1}}_{c_t} + \pi \frac{\phi_t}{2} = \pi c_t + \pi \frac{\phi_t}{2}$$

$$p(0) = \cos^2(\gamma) = \cos^2(\pi\phi_t/2)$$

If $\phi_t = 0 \rightarrow p(0) = 1$ and if $\phi_t = 1 \rightarrow p(1) = 1$

In short, measuring the ancilla gives ϕ_t with probability 1

Iterative Phase Estimation



Subsequent iteration ($t < k$) we use the values of the bits $\phi_t \dots \phi_{k+1}$ found in the previous iterations to set

$$\omega_k = -2\pi \sum_{j=k+1}^t \frac{\phi_j}{2^j} 2^{k-1}$$

Accordingly

$$\begin{aligned} \gamma &= \frac{\omega_k}{2} + \pi \phi 2^{k-1} = \pi \left(- \sum_{j=k+1}^t \frac{\phi_j}{2^j} 2^{k-1} + \sum_{j=1}^t \frac{\phi_j}{2^j} 2^{k-1} \right) \\ &= \pi \sum_{j=1}^k \frac{\phi_j}{2^j} 2^{k-1} = \pi \underbrace{\left(\sum_{j=1}^{k-1} \frac{\phi_j}{2^j} 2^{k-1} \right)}_{c_k} + \pi \frac{\phi_k}{2} = \pi c_k + \pi \frac{\phi_k}{2} \end{aligned}$$

$$p(0) = \cos^2 \gamma = \cos^2(\pi \phi_k / 2) = \begin{cases} 0 \% & \phi_k = 1 \\ 100 \% & \phi_k = 0 \end{cases}$$

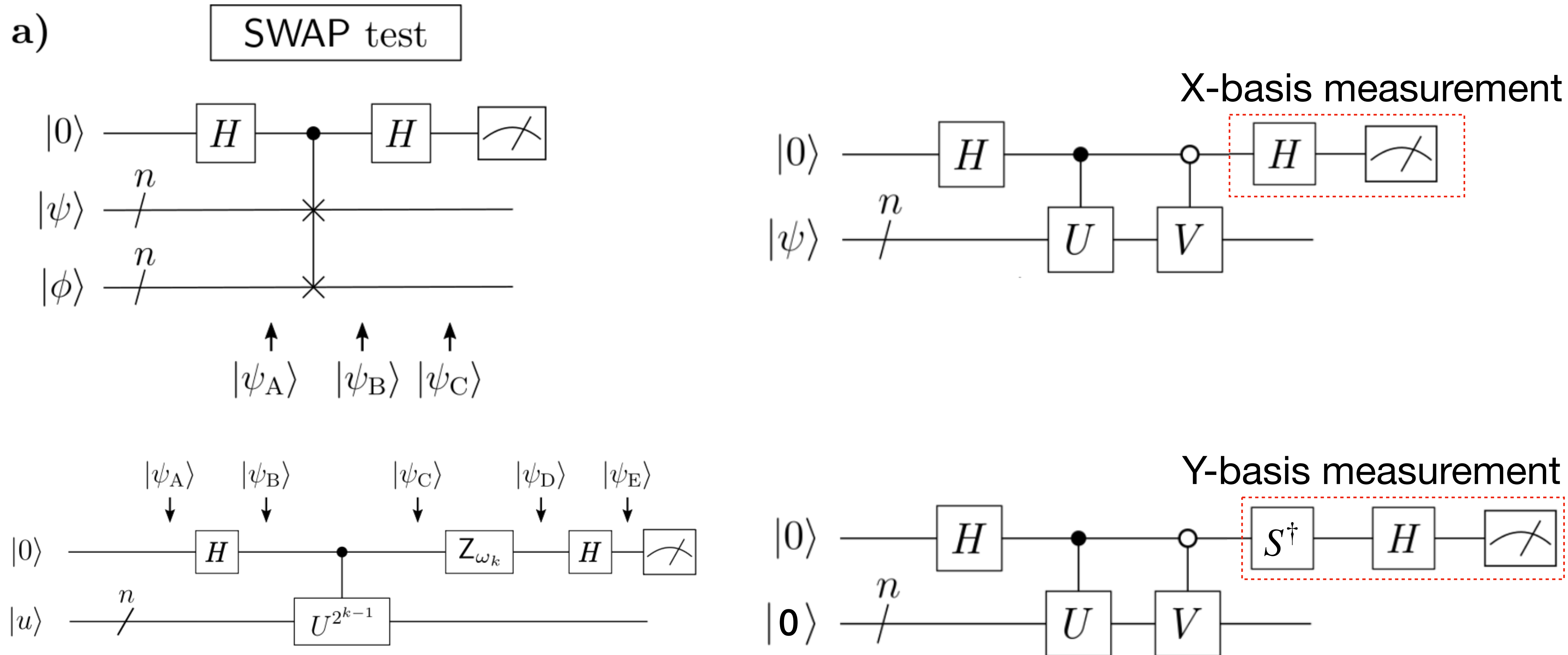
Measuring the ancilla reveals the bit ϕ_k

RPMBT XXIII — Quantum Simulation School

- SWAP test
- Hadamard test
- Iterative phase estimation
- Impact of decoherence
- Bypassing the Hadamard test
 - symmetries
 - mid-circuit measurements

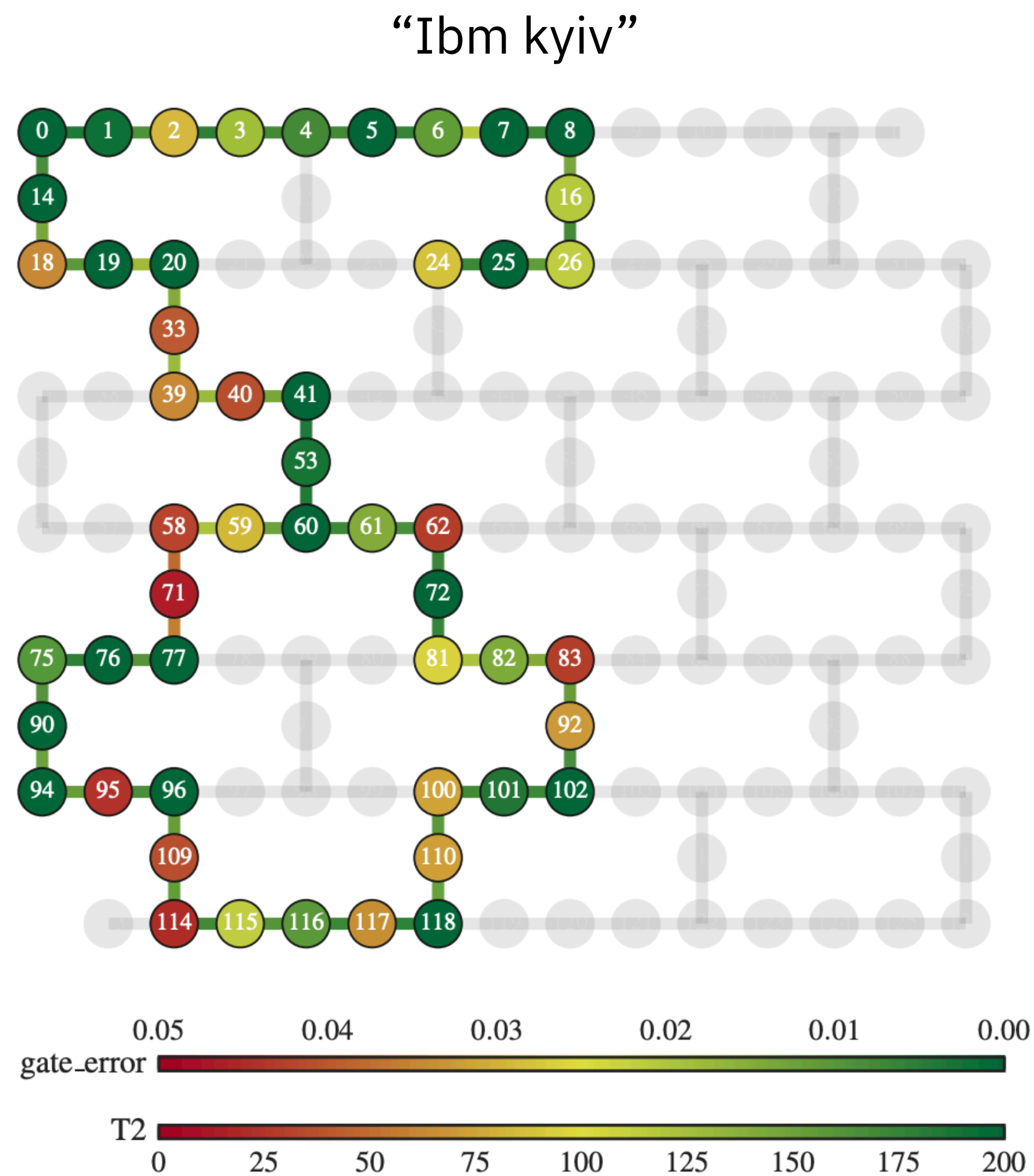
Decoherence

Common features: an ancilla and a deep circuit (many layers of gates)



Decoherence

Common features: an ancilla and a deep circuit (many layers of gates)



- Qubit decoherence (population decay, dephasing)

Depth (number of layers of gates) below

$$D \simeq T_2/t_g \simeq \frac{100 \mu s}{200 ns} = 500$$

Qubit dephasing

Pure dephasing: fluctuations in the qubit frequency, controlled by a “hidden variable” x

$$H_q[x] = -\frac{\omega_q + \delta\omega_q(x)}{2}Z$$

Average over x leads to loss of quantum coherence

$$\rho(t) = \int dx p(x) e^{-itH_q[x]} \rho(0) e^{itH_q[x]} = \rho_{00} |0\rangle\langle 0| + \rho_{11} |1\rangle\langle 1| + \int dx p(x) e^{-it(\omega_q + \delta\omega_q[x])} \rho_{01} |0\rangle\langle 1| + e^{it(\omega_q + \delta\omega_q[x])} \rho_{10} |1\rangle\langle 0|$$

For Gaussian fluctuations, one can prove that

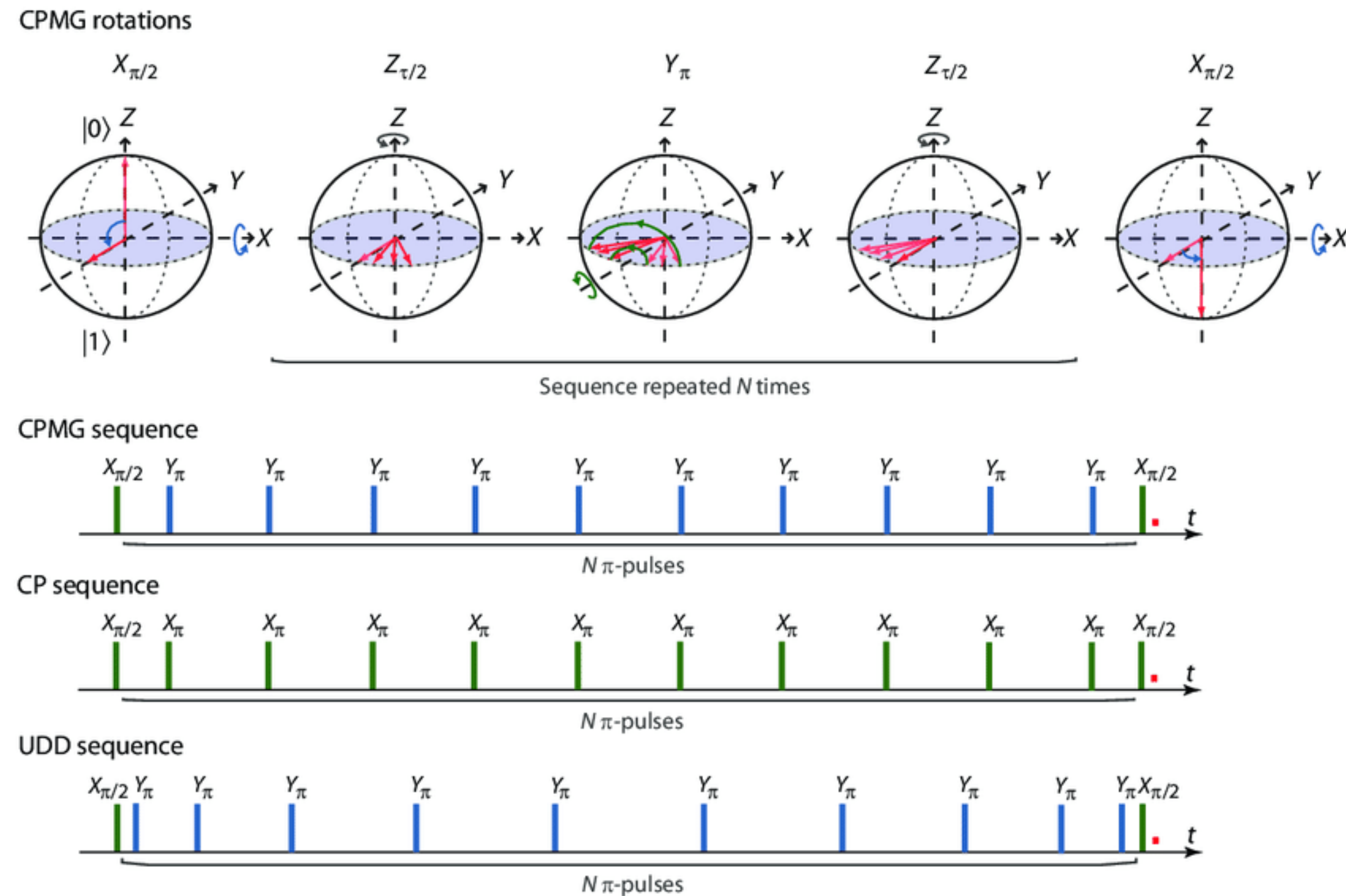
$$\rho(t) = \rho_{00} |0\rangle\langle 0| + \rho_{11} |1\rangle\langle 1| + e^{-\gamma(t)} e^{i\omega_q t} \rho_{01} |0\rangle\langle 1| + e^{-\gamma(t)} e^{-i\omega_q t} \rho_{10} |1\rangle\langle 0|$$

Off-diagonal matrix elements decay based on a dephasing function $\exp(-\gamma(t))$

The state $|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ decays to a mixed state and Hadamard test is compromised.

Dynamical decoupling

Pure dephasing can be weakened by applying sequences of opposite pulses (CPMG sequences)



Dynamical decoupling (the application of e.g. CPMG sequences) weakens pure dephasing, but not all sources of errors on the ancilla (and main register).

RPMBT XXIII — Quantum Simulation School

- SWAP test
- Hadamard test
- Iterative phase estimation
- Impact of decoherence
- Bypassing the Hadamard test
 - symmetries
 - mid-circuit measurements

Bypassing the Hadamard test — symmetries

Suppose U has a symmetry S and one can prepare a “reference state” such that

$$S|R\rangle = s_R|R\rangle \quad , \quad S|\psi\rangle = s_\psi|\psi\rangle \quad , \quad s_\psi \neq s_R \rightarrow \langle\psi|R\rangle = 0 \text{ (different symmetry sector)}$$
$$U|R\rangle = |R\rangle \text{ (trivial evolution)}$$

Then, the quantity $\langle\psi|U|\psi\rangle = \rho e^{i\theta}$ can be estimated from $|\langle\psi|U|\psi\rangle|^2 = \rho^2$ and

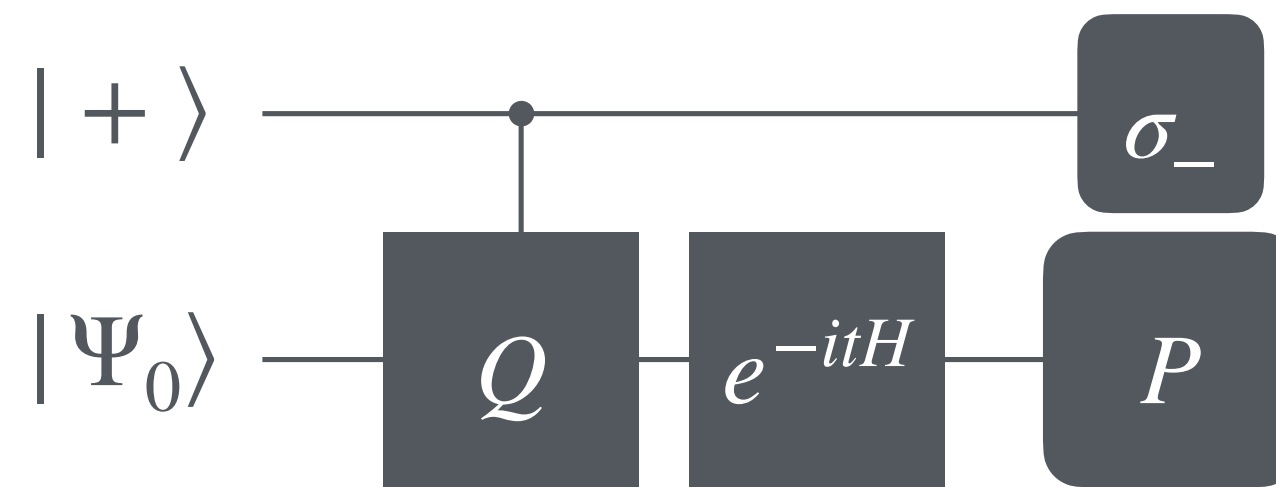
$$F = \left| \frac{\langle R| + \langle\psi|}{\sqrt{2}} U \frac{|R\rangle + |\psi\rangle}{\sqrt{2}} \right|^2 = \left| \frac{\langle R|U|R\rangle + \langle R|U|\psi\rangle + \langle\psi|U|R\rangle + \langle\psi|U|\psi\rangle}{2} \right|^2$$
$$= \left| \frac{1 + \langle\psi|U|\psi\rangle}{2} \right|^2 = \frac{1 + \rho^2 + 2\rho \cos(\theta)}{4} \rightarrow \theta = \arccos \left[\frac{4F - 1 - \rho^2}{2\rho} \right]$$

Bypassing the Hadamard test — circuit cutting

We want to compute a correlation function between Pauli operators P, Q

$$f(t) = \langle \Psi_0 | e^{itH} P e^{-itH} Q | \Psi_0 \rangle \dots$$

but we cannot afford the following “Hadamard test” circuit. What can we do?



We observe that

$$f(t) = \langle \Psi_0 | e^{itH} P e^{-itH} Q | \Psi_0 \rangle = \text{Tr} [P e^{-itH} Q | \Psi_0 \rangle \langle \Psi_0 | e^{itH}]$$

So then, if we could apply the left-multiplication operator $Q | \Psi_0 \rangle \langle \Psi_0 |$ we could bypass the Hadamard test.

But how can we apply a left-multiplication operator?

Bypassing the Hadamard test — circuit cutting

Given a Pauli operator P and a density operator ρ let us define the following operators

$$M_{\pm}[\rho] = \frac{1 \pm P}{2} \rho \frac{1 \pm P}{2}$$

$$R_{\pm}[\rho] = \frac{1 \mp iP}{2} \rho \frac{1 \pm iP}{2}$$

How to physically implement these operators? The “R” operators are simply unitary transformations

$$R_{\pm}[\rho] = \frac{1}{2} \frac{1 \mp iP}{\sqrt{2}} \rho \frac{1 \pm iP}{\sqrt{2}} = \frac{1}{2} e^{\mp i \frac{\pi}{4} P} \rho e^{\pm i \frac{\pi}{4} P}$$

The “M” operators are measurement and post-selection operators: define the projector $\Pi_{\pm} = \frac{1 \pm P}{2}$ and

$$M_{\pm}[\rho] = \Pi_{\pm} \rho \Pi_{\pm} = \text{Tr}[\Pi_{\pm} \rho] \frac{\rho_{\pm}}{\Pi_{\pm} \rho \Pi_{\pm}}$$

- $\text{Tr}[\Pi_{\pm} \rho]$ probability to obtain ± 1
- $\frac{\rho_{\pm}}{\Pi_{\pm} \rho \Pi_{\pm}}$ state after measurement and collapse

Bypassing the Hadamard test — circuit cutting

Given a Pauli operator P and a density operator ρ let us define the following operators

$$M_{\pm}[\rho] = \frac{1 \pm P}{2} \rho \frac{1 \pm P}{2} \text{ and } R_{\pm}[\rho] = \frac{1 \mp iP}{2} \rho \frac{1 \pm iP}{2} \rightarrow Q_k[\rho] = \frac{1 + (-i)^k P}{2} \rho \frac{1 + i^k P}{2}, k = 0, 1, 2, 3$$

These four operators can be arranged into linear combinations to obtain other operators, e.g.

$$\sum_{k=0}^3 i^k Q_k[\rho] = P\rho, \quad \sum_{k=0}^3 (-i)^k Q_k[\rho] = \rho P \quad \text{right-left multiplication operators}$$

$$\sum_{k=0}^3 [i^k + (-i)^k] Q_k[\rho] = \{P, \rho\}, \quad \sum_{k=0}^3 [i^k - (-i)^k] Q_k[\rho] = [P, \rho] \quad \text{commutator and anti commutator operators}$$

Circuit cutting in general

Case 1: circuit cutting [<https://arxiv.org/pdf/1909.07534>]

We are given a bipartite quantum system in $\rho \otimes \sigma$, and we want to apply $e^{i\varphi P \otimes Q}$ (with P, Q Pauli operators)... but we can't afford to apply operations across the partition. What can we do?

We observe that

$$e^{i\varphi P \otimes Q} \rho \otimes \sigma e^{-i\varphi P \otimes Q} = \cos^2 \varphi \rho \otimes \sigma + \sin^2 \varphi P \rho P \otimes Q \sigma Q + i \cos \varphi \sin \varphi [P \otimes Q, \rho \otimes \sigma]$$

$$e^{i\varphi P \otimes Q} \rho \otimes \sigma e^{-i\varphi P \otimes Q} = \cos^2 \varphi [\rho \otimes \sigma] + \sin^2 \varphi [P \rho P \otimes Q \sigma Q] + i \cos \varphi \sin \varphi \frac{[P, \rho] \otimes \{Q, \sigma\} + \{P, \rho\} \otimes [Q, \sigma]}{2}$$

So then, we can represent $e^{i\varphi P \otimes Q} \rho \otimes \sigma e^{-i\varphi P \otimes Q}$ as a linear combination of

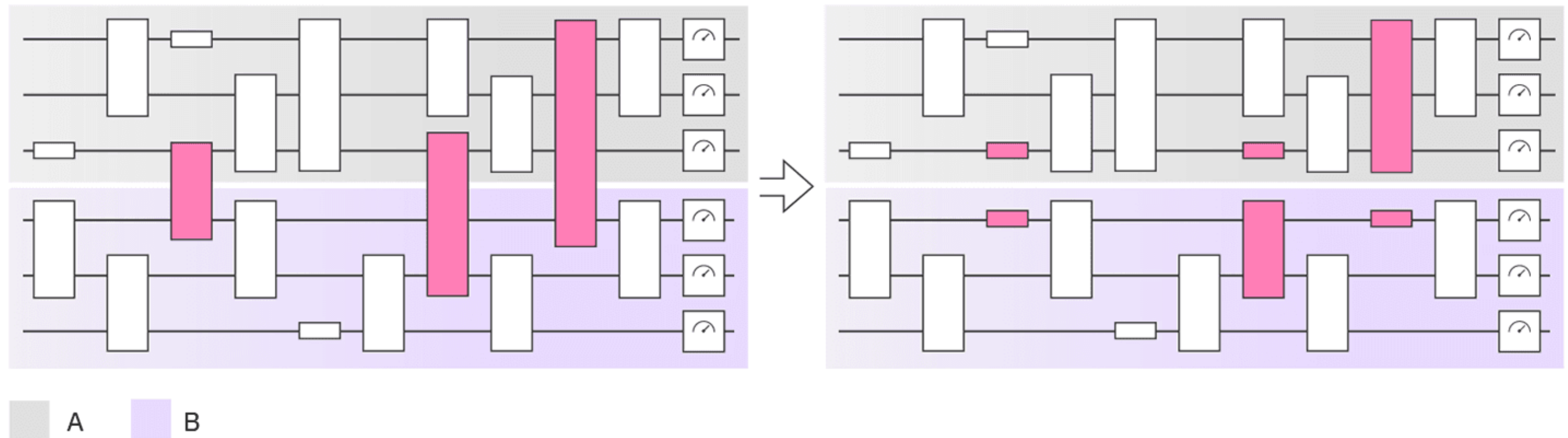
- $\rho \otimes \sigma$ (do nothing)
- $P \rho P \otimes Q \sigma Q$ (application of Pauli operators)
- $[P, \rho] \otimes \{Q, \sigma\} + \{P, \rho\} \otimes [Q, \sigma]$ (application of commutators and anti commutators)

Tensor products = local operations; (anti)commutators = mid-circuit measurements and rotations

Circuit cutting in general

Case 1: circuit cutting [<https://arxiv.org/pdf/1909.07534>]

We are given a bipartite quantum system in $\rho \otimes \sigma$, and we want to apply $e^{i\varphi P \otimes Q}$ (with P, Q Pauli operators)... but we can't afford to apply operations across the partition. What can we do?



Tensor products = local operations; (anti)commutators = mid-circuit measurements and rotations